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October 19, 1961
SUPERSEDING
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April 24, 1952

FEDERAL SPECIFICATION

TAPE, TEXTILE, COTTON, BIAS-CUT

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for cotton, bias-cut tape used primarily for the binding of seams in wearing apparel.

1.2 Classification.

1.2.1 *Types and classes.* The bias-cut cotton tape shall be furnished in the following types and classes, as specified (see 6.2).

Type I.—Cambric.
Type II.—Percale.
Type III.—Sateen.
Type IV.—Twill.
Class 1.—Bleached.
Class 2.—Dyed.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

V–T–276—Thread, Cotton.
UU–P–268—Paper, Kraft, Untreated, Wrapping.
CCC–T–191—Textile Test Methods.
PPP–B–636—Box, Fiberboard.
PPP–T–60—Tape; Pressure Sensitive Adhesive, Waterproof—For Packaging and Sealing.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Std. No. 751—Stitches, Seams, and Stitching.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific

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procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

Rules and Regulations Under The Textile Fiber Products Identification Act.

(Copies may be obtained, free of charge, from the Federal Trade Commission, Washington 25, D. C.)

Color Association of the United States, Inc.

Standard Color Card of America

or

U. S. Army Standard Color Card for the Official Colors for Arms and Services.

(Application for copies should be addressed to the Color Association of the United States, Inc., 200 Madison Avenue, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is required, the tape shall match that standard sample for color and shall be equal to, or better than, the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.

3.2.1 *Cloth.* The cloth to be cut into bias tape shall be cotton and shall conform to the requirements shown in table I for the type tape specified when tested as specified in 4.4. In addition, type I cloth shall contain sufficient sizing to produce a cambric finish.

TABLE I.—*Physical requirements*

Type	Weight per square yard (minimum)	Yarns per inch (minimum)		Weave
	Ounces	*Warp*	*Filling*	
I	2.7	64	55	Plain
II	3.0	85	72	Plain
III	2.8	88	140	Sateen
IV	4.0	76	114	½ Twill

3.3 Color. The cloth shall be bleached or dyed the color specified (see 6.2). The use of sulfur dyes or substances containing elementary sulfur compounds capable of oxidation to sulfuric acid is prohibited. Test for sulfur dyes shall be made as specified in 4.4. Unless otherwise specified, when ordering red, blue, or green colors all of the following numbers in each series of red, blue, and green cable numbers of the Standard Color Card of America are acceptable and may be furnished at the option of the supplier.

Red	*Blue*	*Green*
No. 70180	No. 70216	No. 70167
No. 70081	No. 70085	No. 70168
	No. 70086	

3.3.1 *Matching.* The color shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard shade under incandescent lamplight at 2800° Kelvin.

3.3.2 *Colorfastness.* The finished tape shall show fastness to laundering, bleaching, wet-dry cleaning, crocking, light, and perspiration equal to or better than the standard shade sample. When no standard sample is available, the finished tape shall show "good" fastness to laundering, bleaching, wet-dry cleaning, crocking, light and perspiration. Tests shall be made as specified in 4.4.

3.4 Thread. The thread shall be cotton, type IA2 or IA3, 3 ply, ticket No. 60, conforming to Federal Specification V–T–276. The thread shall be bleached or dyed to approximate the shade of the tape, as specified. When dyed, the thread shall show fastness to laundering, bleaching, wet-dry cleaning, light and perspiration equal to, or better than, the standard sample. When no standard sample is available, the thread shall show good fastness to laundering, bleaching, wet-dry cleaning, light, and perspiration.

3.5 Construction. The cloth to be cut into bias tape shall have the selvages removed and then shall be cut on an angle of 45° ± 5°. The tape shall then be joined together. The seam joining shall be securely sewn with not less than

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10 nor more than 14 stitches per inch, using stitch type 301 and seam type SSa–1 of Federal Standard No. 751 with seams pressed open. Types II and IV shall have the edges of the tape treated with sizing to prevent raveling. The sizing shall not affect the colorfastness requirements of the finished tape provided in 3.3.2.

3.6 Width. The width of the tape shall be as specified (see 6.2). A tolerance of $\pm 1/32$ of an inch will be permitted.

3.7 Length. The tape shall be rolled on tubes of the same width as the tape. Unless otherwise specified, the length of the rolls shall be not less than 50 nor more than 100 yards, as specified (see 6.2).

3.8 Fiber content. The tape shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations Under The Textile Fiber Products Identification Act, effective March 3, 1960.

3.9 Workmanship. The finished tape shall conform to the quality and grade of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Contractor's inspection responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection for acceptance. Except when otherwise specified by civil government agencies, inspection shall be in accordance with Military Standard MIL–STD–105 except where otherwise indicated.

4.2.1 *Inspection of component.* Inspection of the thread shall be performed in accordance with the quality assurance provisions of the subsidiary specification except that the inspection level shall be L2, table III-B of Military Standard MIL–STD–105.

4.2.2 *Examination of the end item.* Defects found during the examination shall be classified in accordance with 4.2.2.1, 4.2.2.2, and 4.2.2.3.

4.2.2.1 *Yard-by-yard examination.* The required yardage of each roll shall be inspected for visual defects listed below. All defects found shall be counted, regardless of their proximity one to another, except when two or more defects represent a single local condition in which case only the more serious defect will be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The unit of product for this examination shall be 1 linear yard. Except when otherwise specified by civil government agencies, the sample size shall be in accordance with inspection level I of Military Standard MIL–STD–105. The acceptable quality level (AQL) shall be 4.0 defects per 100 units (yards). The lot size shall be expressed in units of 1 yard each. An approximate equal number of yards shall be examined from each roll selected. The number of rolls from which the sample is to be selected shall be in accordance with table II of this specification.

	Defects
Material defects	Holes, cuts, tears, multiple floats, smash—Any.
Abrasion marks	Resulting in a weak place.
Crease or wrinkle	Permanently embedded.
Stain, spot or streak	Single, more than 1 inch in combined direction or multiple small spots within 9 linear inches—Clearly visible at normal inspection distance (approximately 3 feet).
Selvage	Not removed.
Seams and stitching	Any open seam. Stitches per inch more or less than specified. Insecurely stitched. Seams not pressed open. Wrong stitch or seam type.

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NOTE: A seam shall be classified open when one or more stitches joining a seam are broken or when two or more continuous skipped stitches occur.

Stitch tension	Loose tension resulting in a loosely secured seam.
	Tight tension—Stitches break when normal strain is applied to seam or stitching.

NOTE: Puckering is evidence of tight tension. When evident, test seam by exerting moderate pull on seam.

Width	Not within specified tolerances.
Type I only	Finish not equal to standard.
Angle of bias	Less than 40 degrees.
	More than 50 degrees.

TABLE II.

Lot size in yards	Sample size in rolls	Maximum number of defects acceptable in sample
Up to 1,300	3	0
1,301 to 3,200	5	0
3,201 to 8,000	7	0
8,001 to 22,000	10	0
22,001 to 110,000 ..	15	0
110,001 and over ..	25	1

4.2.2.2 *Overall examination.* Each defect listed below shall be counted not more than once in each roll examined. The unit of product for this examination shall be one roll. The sample size (same number of pieces as the yard-by-yard examination) and acceptance number shall be as shown in table II.

Defects

Off shade, not within established tolerances.

Uneven dyeing.

Poor penetration, mottled or streaky throughout.

Objectionable odor.

Overall uncleanness.

4.2.2.3 *Length examination.*

4.2.2.3.1 *Examination for length of individual rolls.* Each roll in the sample shall be examined for gross length. Any gross length found to be less than the specified minimum length or more than the maximum length permitted, or any gross length found to be more than 2 yards less than the gross length marked on the ticket, shall be considered a defect with respect to length. The unit of product for this examination shall be one roll. The sample size and acceptance number shall be as shown in table II.

4.2.2.3.2 *Examination for total yardage in sample.* The lot shall be unacceptable if the total of the actual gross length of rolls in the sample is less than the total of the gross lengths marked on roll tickets.

4.3 Examination of preparation for delivery requirements. An examination shall be made to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be L–4 and the AQL shall be 4.0 defects per hundred units.

Examination	*Defect*
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as: Incomplete closure of container flaps, improper taping, loose strapping, inadequate stapling. Bulging or distortion of containers.
Weight	Net weight exceeds requirements.

4.4 Testing. The methods of testing specified in Federal Specification CCC–T–191, wherever applicable, and as listed in table III shall be followed. The physical and chemical values

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specified in section 3 apply to the determinations made on a unit of product for test purposes as specified in the applicable test method. The lot size shall be expressed in units of one linear yard of tape, full width before cutting. The unit of product for test purposes shall be two linear yards, full width. The lot shall be unacceptable if one or more units fail to meet any requirement specified. The sample size shall be as follows:

Lot size	*Sample size*
800 or less	2
801 to 22,000	3
22,001 and over	5

TABLE III.—*Test methods*

Characteristic	Requirement reference	Test method
Weave[1]	Table I	Visual
Yarns per inch	Table I	5050
Weight; ounces per square yard	Table I	5041
Identification of cotton	3.2.1	1200
Colorfastness to:		
Laundering	3.3.2	5610
Bleaching	3.3.2	5600
Wet-dry cleaning	3.3.2	5622
Crocking	3.3.2	5651
Light[2]	3.3.2	5660
Perspiration[3]	3.3.2	5680 or 5682
Sulfur[1]	3.3	(see 4.4.1)
Width of binding	3.6	5020
Length of rolls	3.7	5010

[1] One determination per unit of product and results reported as pass or fail.

[2] In the event of a dispute resulting from the test with method 5660 or as a result of suspected anomalous behavior of certain dye types or formulations, the contracting officer may authorize the exposure to natural light in accordance with method 5662.

[3] The transfer cloth shall consist of a warp of polyester fiber or Orlon 81 and a 6-fiber filling; with equal bars of wool, spun nylon, spun rayon, spun silk, spun acetate and bleached cotton. Each 6-fiber repeat shall measure approximately 2 inches and shall be separated by a waste filling stripe.

4.4.1 *Test for presence of sulfur.* Presence of sulfur is determined as follows: Dissolve 100 grams of stannous chloride crystals in 100 ml. hydrochloric acid (35 percent concentration) and add 50 ml. of distilled water. To preclude interference in dye identification, sample first shall be boiled in distilled water to remove any trace of chemical used in the dyeing process. One to 1.5 grams of the finished tape, cut into small pieces shall be used for the test. Place the sample in a test tube of suitable size and cover with stannous chloride solution. Place a filter paper which has been moistened with one drop of lead-acetate solution (made by dissolving 5 grams of reagent-grade lead acetate to 100 cc. of distilled water; add a drop of glacial acetic acid to clear up if cloudy) over the mouth of the test tube. (Punch needle hole in paper.) Heat slowly over a low flame until the solution is boiling. Solution shall not be boiled for more than 10 to 15 seconds as prolonged boiling will reduce the sensitivity of the test. A brown to black stain appearing on the paper indicates the presence of sulfur.

5. PREPARATION FOR DELIVERY

(Federal Standard No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civil agency procurement.)

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A.* Tubes of tape, not exceeding multiples of 10, of one type, class, and width only shall be wrapped together with 30-pound minimum basis weight kraft paper conforming to grade B of Federal Specification UU–P–268. The wrap shall completely enclose the tape and be secured with gummed or pressure-sensitive tape.

5.1.2 *Level C.* Tape shall be packaged in accordance with the industry's practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 *Level A.* Tape of one type, class, and width only, packaged as specified in 5.1.1, shall be packed in a snug-fitting fiberboard shipping container conforming to class 2, V3c or V3s, style RSC–L of Federal Specification PPP–B–636. The inside of each shipping container shall be

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fitted with a one-piece double-faced corrugated liner having a minimum bursting strength of 275 p.s.i. conforming to Federal Specification PPP–B–636. The ends of the liner shall abut at the opposite corner from the manufacturer's joint. Each container shall be closed and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification. In addition, all seams, corners, and manufacturer's joint of the container shall be sealed by the application of a 3-inch-minimum width cloth tape conforming to type III, class 1, olive drab or other acceptable color of Federal Specification PPP–T–60. The net weight of the contents of each shipping container shall not exceed 65 pounds.

5.2.2 *Level B.* Tape of one type, class, and width only, packaged as specified in 5.1.1, shall be packed in a snug-fitting fiberboard shipping container conforming to type I, or II, class 1, style RSC–L of Federal Specification PPP–B–636, except that the minimum bursting strength of the fiberboard shall be 275 p.s.i. The inside of each shipping container shall be fitted with a one-piece double-faced corrugated liner having a minimum bursting strength of 275 p.s.i. conforming to Federal Specification PPP–B–636. The ends of the liner shall abut at the opposite corner from the manufacturer's joint. All flaps of the fiberboard shipping containers shall be securely sealed with an adhesive commercially used for the specific product being packed by application throughout the entire area of contact between the flaps, or by a combination of metal stitching the bottom flaps and sealing the top flaps with adhesive. The net weight of the contents of each shipping container shall not exceed 65 pounds.

5.2.3 *Level C.* Tape, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with rules and regulations of carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Intended use. The cotton bias-cut tape covered by this specification is intended primarily for the binding of seams in wearing apparel.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type and class of tape required (see 1.2.1).
(*c*) Color required (see 3.3).
(*d*) Length of rolls (see 3.7).
(*e*) Width of tape (see 3.6).
(*f*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Sample. For access to standard sample for color, address the procuring office issuing the invitation for bids.

6.4 Transportation description. The transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Tape, cotton, not otherwise indexed by name.
Carload minimum weight 20,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Tape, not otherwise indexed (in packaged units of not less than 24 continuous yards per unit).
Truckload minimum weight (W) 24.6 pounds, (W) subject to Rule 34, National Motor Freight Classification.

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6.5 Level A packaging and level B packing shall be used by: ordering agencies only when shipment is known to be made to a holding supply depot or warehouse from which partial shipments will be made to ultimate consumers.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:

Army—M Q
Navy—S MC
Air Force—MAA

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